Minseo Kim

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Research Interests

AI Systems, Efficient Inference across Modern Architectures and Workloads

Education

- **University of California, Berkeley** — *Aug 2026 – Present*
  Ph.D. in Computer Science — Berkeley, CA, USA
  ◦ Advisor: Prof. Kurt Keutzer
- **Seoul National University** — *Mar 2020 – Aug 2026*
  *B.S. in Computer Science and Engineering, **Summa cum laude** (GPA: 4.03/4.30)* — Seoul, South Korea
  ◦ Military service in the Republic of Korea Air Force (Oct 2021 – Jul 2023)
- **University of California, Berkeley** — *Jun 2025 – Aug 2025*
  Summer Session Visiting Student (GPA: 4.0/4.0) — Berkeley, CA, USA
- **Korea Minjok Leadership Academy** — *Mar 2017 – Feb 2020*
  *Secondary Education, **First-class honors** in STEM (1st / 73)* — Gangwon-do, South Korea

Research Experience

- **Berkeley Artificial Intelligence Research (BAIR) Lab [🌐]** — *Aug 2026 – Present; Jun 2025 – Jan 2026*
  Ph.D. Student (Advisor: Prof. Kurt Keutzer) — Berkeley, CA, USA
  Undergraduate Researcher
- **SNU Architecture and Code Optimization Lab [🌐]** — *Jan 2025 – Feb 2025*
  Undergraduate Researcher (Advisor: Prof. Jae W. Lee) — Seoul, South Korea

Work Experience

- **FuriosaAI [🌐]** — *Jan 2026 – Jul 2026*
  AI Algorithm Research Intern — Seoul, South Korea
- **Deeping Source Inc. [🌐]** — *Jul 2024 – Aug 2024*
  ML Engineering Intern — Seoul, South Korea

Preprints

* Equal contribution † Co-corresponding authors

[3] MetaCurate: Evolving Curation Filters for High-Quality Agentic Coding Data. [Paper] [Code]
Donghoon Kim*, Seunghyuk Oh*, Kevin Galim, **Minseo Kim**, Minjae Lee, Younghyun Kim, Wooseok Seo, Hyung Il Koo, Wonjun Kang
Preprint, 2026

[2] Elicitation or Expansion? Demystifying Gains from Agent Harness Optimization. [Paper] [Code]
Younghyun Kim*, Wooseok Seo*, **Minseo Kim**, Seunghyuk Oh, Minjae Lee, Donghoon Kim, Hyung Il Koo, Jaehyung Kim†, Wonjun Kang†
Preprint, 2026

[1] Beyond Next-Token Prediction: A Performance Characterization of Diffusion versus Autoregressive Language Models. [Paper]
Minseo Kim, Coleman Hooper, Aditya Tomar, Chenfeng Xu, Mehrdad Farajtabar, Michael W. Mahoney, Kurt Keutzer, Amir Gholami
Preprint, 2025

Publications & Workshop Papers

[9] SketchSSM: Write to the Full State, Read from a Compact Sketch. [Paper]
Omin Kwon, JoongWon Shin, **Minseo Kim**, Kurt Keutzer, Sehoon Kim†, Jae W. Lee†
NeurIPS 2026 LCFM Workshop

[8] EfficientRollout: System-Aware Self-Speculative Decoding for RL Rollouts. [Paper] [Code]
Minseo Kim*, Minjae Lee*, Seunghyuk Oh, Kevin Galim, Donghoon Kim, Coleman Hooper, Harman Singh, Amir Gholami, Hyung Il Koo, Wonjun Kang
NeurIPS 2026
ICML 2026 AdaptFM Workshop

[7] AsyncOPD: How Stale Can On-Policy Distillation Be? [Paper] [Code]
Wonjun Kang*, Kevin Galim*, Seunghyuk Oh, Minjun Kang, Sanghyun Park, Donghoon Kim, Minjae Lee, **Minseo Kim**, Rishabh Tiwari, Yuchen Zeng, Hyung Il Koo, Kangwook Lee
NeurIPS 2026
ICML 2026 DEMO Workshop

[6] MAGE: All-[MASK] Block Already Knows Where to Look in Block Diffusion LLM. [Paper]
Omin Kwon, Yeonjae Kim, Doyeon Kim, **Minseo Kim**, Yeonhong Park, Jae W. Lee
NeurIPS 2026
ICML 2026 AdaptFM Workshop **Best Paper Award**

[5] LLM Inference in a Flash! [Paper]
Sebastian Zhao*, **Minseo Kim***, Coleman Hooper*, Luca Manolache, Michael W. Mahoney, Yakun Sophia Shao, Kurt Keutzer, Amir Gholami
ISCA 2026 MLArchSys Workshop **Oral**

[4] CATS: Inference-aligned SFT for Diffusion LLMs via Context-sensitivity Aware Trajectory Sampling. [Paper]
Seunghyuk Oh, Minjae Lee, Kevin Galim, **Minseo Kim**, Hyung Il Koo, Wonjun Kang, Hanbaek Lyu, Kangwook Lee
ICLR 2026 DeLTa Workshop

[3] CDLM: Consistency Diffusion Language Models for Faster Sampling. [Paper] [Code]
Minseo Kim, Chenfeng Xu, Coleman Hooper, Harman Singh, Ben Athiwaratkun, Ce Zhang, Kurt Keutzer, Amir Gholami
MLSys 2026

[2] KRETA: A Benchmark for Korean Reading and Reasoning in Text-Rich VQA Attuned to Diverse Visual Contexts. [Paper] [Code] [Dataset]
Taebaek Hwang*, **Minseo Kim***, Gisang Lee, Seonuk Kim, Hyunjun Eun
EMNLP 2025

[1] InstANNS: Scalable Approximate Nearest Neighbor Search via Cost-Efficient In-Storage Processing. [Paper]
Bonggeun Sim, Yushin Kim, **Minseo Kim**, Yeonhong Park, Jae W. Lee
CIKM 2025

Awards & Achievements

- **NIPA President's Award (Grand Prize) – 2025 AI Chip Contest** *Dec 2025*
- **Ministry of Science and ICT Award – Whitehat School Outstanding Graduate** (1st / 309) *Mar 2024*
- **Vulnerability Disclosures** (CVE-2024-11510, CVE-2024-11511, CVE-2024-11512) *2023*

Scholarships

- **Gaheonsindo Foundation Scholarship** (Full Undergraduate Tuition) *2024 – 2026*
- **SNU Specialized Semiconductor College Scholarship** *2024 – 2026*

Talks & Leadership

- **Diffusion Language Model Seminar @ Cerebras** (Virtual) *Jan 2026*
- **President @ Guardian** (SNU CSE Security Research Club) *Aug 2024 – Aug 2025*
- **Windows SW Vulnerability Analysis Seminar @ 28th Winter Hacking Camp** *Feb 2024*

Language Proficiency

Korean (Native), English (Fluent)

References

Prof. Kurt Keutzer — University of California, Berkeley (Ph.D. Advisor)
Dr. Wonjun Kang — FuriosaAI (Internship Mentor)
Prof. Jae W. Lee — Seoul National University (Undergraduate Advisor)